

AI VISION EXTRACT

Automated Subject Isolation using Deep Learning

Semantic Image Segmentation with DeepLabV3

PROJECT OBJECTIVE

The primary goal is to build a robust Machine Learning model capable of automatic subject extraction from images. By utilizing Deep Learning, we aim to achieve high-precision semantic segmentation.

CORE CHALLENGE

Achieving accurate pixel-wise classification to seamlessly separate complex foreground subjects from diverse backgrounds.

Key Specifications

-  **Input:** Any RGB image containing a discernible subject and background context.
-  **Output:** Subject fully isolated on a pure black or transparent background.
-  **Applications:** Photography automation, virtual conferencing, AR/VR integration, and background replacement.
-  **Method:** DeepLabV3 architecture utilizing a pre-trained ResNet101 backbone.

LEARNING OUTCOMES

- ✓ **Semantic Segmentation:** Mastered principles of pixel-wise classification and deep learning architectures like DeepLabV3.
- ✓ **Data Pipeline:** Implemented preprocessing workflows for mask annotations and pixel-wise tasks.
- ✓ **Model Evaluation:** Utilized IoU and Dice coefficient metrics for rigorous model validation.
- ✓ **Deployment:** Built a production-ready solution with a responsive Streamlit web application.
- ✓ **MLOps Mastery:** Executed end-to-end workflow from raw data ingestion to inference API.



COCO 2017 DATASET ANALYSIS & PROCESSING

DATASET OVERVIEW

Large-scale object detection & instance segmentation benchmark serving as the foundation for our model training.

Training Set Statistics

- **118,287** total images in raw dataset
- **117,254** valid images after filtering (99.1% retention)
- **80** distinct object categories with high-quality instance masks

Validation Set

- **5,000** total images available
- **4,952** valid after filtering (99.0%)
- Maintains same 80 categories for consistent evaluation

Processing Pipeline

Implemented robust anomaly detection to handle missing files and corrupted images.

- Minimum box area threshold: **32 pixels** (5.7×5.7)



IMAGE DATA PREPROCESSING

ANOMALY DETECTION PIPELINE

Automated identification of missing files and corrupted images. Filters invalid annotations and eliminates tiny bounding boxes below the 32-pixel threshold to ensure data integrity.

MASK EXTRACTION PROCESS

- Converts instance masks to semantic masks with COCO category IDs (0-80).
- Merges overlapping masks using RLE encoding.
- Saves processed masks as lossless PNG files.

QUALITY ASSURANCE

Achieved >99% valid data retention rate with detailed anomaly reports and binary validation checks for every image in the dataset.



Part 2/2: Image Transformation Pipeline

MODEL INPUT NORMALIZATION

- **Resize Operation:**

Target size of 520x520 pixels to meet ResNet101 backbone requirements while preserving aspect ratio considerations.

- **Tensor Conversion:**

Images are normalized to range [0, 1] and undergo channel-wise normalization using standard ImageNet statistics.

- **Batch Processing:**

Implemented with a batch size of 8 images per iteration, GPU-optimized for Kaggle/Cloud environments.

Normalization Parameters

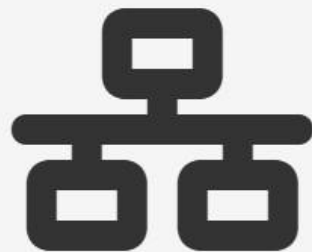
Mean (RGB)	[0.485, 0.456, 0.406]
Std (RGB)	[0.229, 0.224, 0.225]

Rationale:

The pre-trained ResNet backbone expects inputs normalized with specific ImageNet statistics. Failing to match this distribution would significantly degrade transfer learning performance.

DEEPLABV3 RESNET101 ARCHITECTURE

- **Backbone: ResNet101** — A 101-layer residual network pre-trained on ImageNet. Extracts multi-scale features with strides of 4, 8, and 16.
- **ASPP Module** — Atrous Spatial Pyramid Pooling captures multi-scale contextual information using dilated convolutions at different rates.
- **Decoder** — Upsamples feature maps to original resolution. Outputs 81 channels (80 COCO categories + 1 background) for pixel-wise classification.
- **Loss Function** — Utilizes CrossEntropyLoss to handle multi-class segmentation and class imbalance across the dataset.



TRAINING CONFIGURATION

⚙️ Optimization Strategy

Optimizer: SGD (Stochastic Gradient Descent)

Learning Rate: **0.01** with Momentum **0.9**

Regularization: Weight decay **1e-4** (L2) to prevent overfitting

🔄 Training Loop

Forward: DeepLabV3 outputs dictionary with 'out' key

Loss: CrossEntropyLoss vs. ground-truth masks

Backward: Loss tracking & weight updates via SGD

🖨️ Batch & Hardware

Batch Size: **8 images** per iteration

Environment: GPU-optimized for Kaggle/Colab (NVIDIA)

Data Handling: Shuffled training loader, sequential validation

👁️ Inference Mode

Configuration: Model set to **eval()**

Optimization: **torch.no_grad()** disables gradients

Stability: Batch Norm uses running statistics; no dropout

STREAMLIT WEB APPLICATION

INTERACTIVE DEPLOYMENT PLATFORM



USER INTERFACE

Responsive dark-mode design with sidebar settings panel for real-time control adjustments.



IMAGE PROCESSING

Support for single or batch uploads (JPG/PNG). Real-time inference with side-by-side comparison.



CUSTOM BACKGROUNDS

Instant replacement with Pure Black (#000), White (#FFF), or Studio Gray (#2D2D2D).



ENHANCEMENTS

Post-segmentation controls for Brightness (0.5–2.0×) and Contrast adjustment.



ADVANCED FEATURES

Auto-crop to subject, subject coverage metrics (%), and original resolution display.



EXPORT OPTIONS

Download processed images as PNG or batch ZIP with full metadata tags included.



PROJECT OUTPUT & FUTURE SCOPE

Current Achievements

- ✔ **High Performance Metrics:**
Tracked via IoU, Dice Coefficient, and Pixel-wise Accuracy.
- 📁 **Robust Dataset Processing:**
122,206 clean semantic masks extracted with 99%+ quality retention.
- 🔧 **Production Deployment:**
GPU-optimized Streamlit application capable of batch processing.

Future Roadmap

- 🎯 **Domain-Specific Fine-tuning:**
Optimization for e-commerce products, portrait photography, and medical imaging.
- 🎥 **Real-Time Video Processing:**
Frame-wise segmentation with temporal smoothing for live background replacement.
- ⚙️ **Advanced Model Optimization:**
ONNX export, INT8 quantization, and distillation for edge devices.